# Final Project Report

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11/8/2022

## 1 Prelab

Given the transistor datasheets for the 2N3904 and 2N3906 BJTs, we determine the values of the parameters $C_\pi$, $C_\mu$ and other necessary parameters by reading the corresponding graphs. Let us consider the results for both below:

According to the datasheet, we have the following values for the small-signal characteristic $C_{\mu 0} = 3.6\text{pF}$ and $C_{je0} = 4.5\text{pF}$. Then, in order to determine $C_\pi = C_{je} + g_m \tau_f$, we consider the equations $C_{je} = 1.8 C_{je0} = 8.1\text{pF}$ and $C_\pi = C_{je} + \tau_F \cdot g_m$, for some corresponding $g_m$. On the other hand, we have $C_\mu = \frac{1}{\sqrt{1+\frac{V_B}{0.7V}}}$, for some corresponding $V_B$.

## 2 Design

In order to derive the values corresponding to each of the resistors in such a way that they satisfy the constraints requested, we start from the biasing. We wanted our biasing to mainly satisfy two constraints: **1.** the power dissipation had to be lower than 100mW, which implies that the total current going through all the branches could not be larger than 6.67mA; and **2.** the voltage gain for the second stage, corresponding to $A_v = -g_{m2} R_C$ must be greater than at least 100.

First, we know that the current going through transistor Q2 is dependent on the fixed voltage $V_{E2}$ and the resistor $R_{E2}$; thus, we can choose the resistor and the voltage in such a way that we get our desired current. In our case, we chose a current $I_{C2} \approx 4.4\text{mA}$. Following this, we want to set the voltage $V_{E2} = 1.7\text{V}$, which in turn would result in $V_{E1} = 2.4\text{V}$ and $V_{B1} = 3.1\text{V}$. The resistor chosen is $R_{E2} = 390\Omega$.

Once this is given, we set the resistors $R_1 = 5.6\text{k}\Omega$ and $R_2 = 15\text{k}\Omega$ and $R_{E1} = 2.6\text{k}\Omega$ such that $V_{B1} = (15V)\frac{R_2}{R_2+R_1} = 3.1\text{V}$ and to have a small current $I_{C1} = 0.92\text{mA}$. Then, we choose the rest of the biasing as follows:

$$V_{C2} = 2.9V$$

$$V_{C3} = 8.6V$$

$$V_{B3} = 3.6V$$

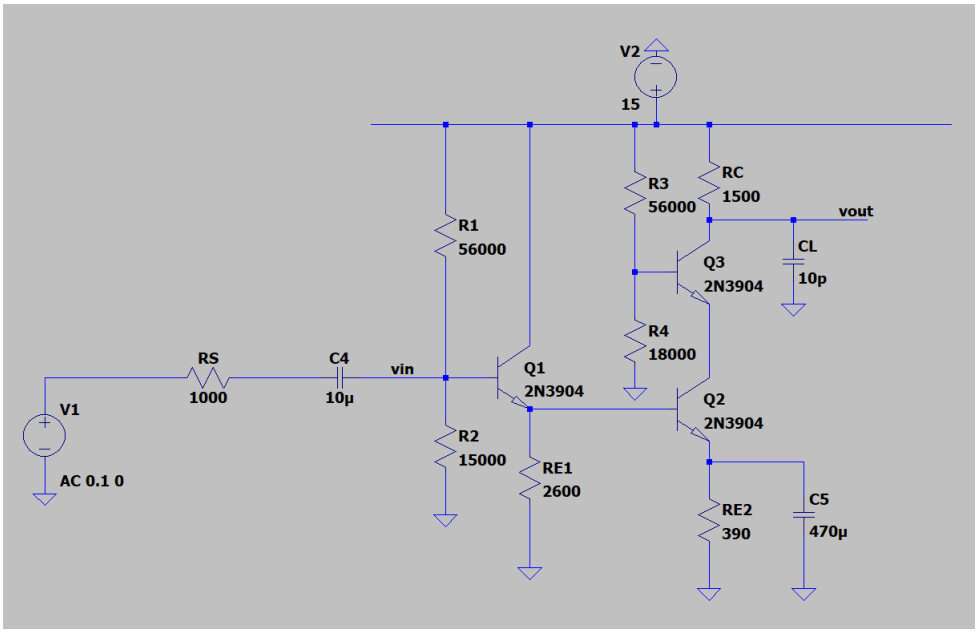



Figure 1: Current topology chosen for the project.

Note that the voltage $V_{C3} = 8.6\text{V}$ is such that $15V - V_{C3} > 2V$ and $V_{C3} - V_{C2} > 2V$, thus allowing for a voltage swing peak-to-peak larger than 2V. In order to satisfy the constrain of $V_{B3} = (15V)\frac{R_4}{R_3+R_4} = 3.6\text{V}$, we chose the resistors $R_3 = 56\text{k}\Omega$ and $R_4 = 18\text{k}\Omega$. The restriction in $V_{C3}$ also requires that $R_C = 1.5\text{k}\Omega$.

Following this, we have that the gain of this amplifier will be given by:

$$A_v = A_{v1} \cdot A_{v2}$$

where $A_{v1}$ and $A_{v2}$ are the first and second stages of amplification, respectively. We can assume that the first stage of amplification will work as an ideal buffer due to its common-collector configuration, thus having that $A_{v1} \approx 1$. Then, for the second stage we will have a gain of $A_{v2} = -g_{m2}R_C$. Given that $g_{m2} = \frac{4.4mA}{26mV} \approx 0.17\text{mS}$, we will end up with a small-signal gain of:

$$A_v = 1 \cdot A_{v2} = (0.17mS)(1500\Omega) = 253.85$$

which is well beyond the requirements of this project. In addition to this, we have that the currents $I_{C2} + I_{C1} < 6.67\text{mA}$, also satisfying the power consumption constraints.

Finally, we consider the bandwidth of the amplifier. In order to do this, we calculate the zero time-value constants for the combined capacitors $C_1 = C_{\pi 2} + 2C_{\mu 2}$, $C_2 = C_{\pi 3}$, and $C_3 = C_L + C_{\mu 3}$. We neglect the values for $C_{cs}$ due

to lack of parameters provided in the datasheets. Looking at the diagram, we obtain:

- $\tau_{C_1} = C_1(R_{E1}||r_{\pi 2}||\frac{1}{g_{m1}})$
- $\tau_{C_2} = C_2(\frac{1}{g_{m3}})$, assuming that $r_{o2} >> \frac{1}{g_{m3}}$
- $\tau_{C_3} = C_3(R_C)$, assuming that $r_{o3} >> R_C$

Plugging in the corresponding values and using $f_h = \frac{\omega_h}{2\pi}$, we obtain:

$$f_h = \frac{1}{2\pi(\tau_{C_1} + \tau_{C_2} + \tau_{C3})} = 5.68\text{MHz}$$

Then, to calculate the lower bandwidth limit, we calculate the infinite time-value constants for the coupling and bypass capacitors sitting at $R_{E2}$ and $R_1$ respectively. We assign them large values (such as $470\mu$F) in order to minimize the value of $f_l$ as much as possible:

- $\tau_{C_4} = C_4(R_{E2}||\frac{1}{g_{m1}})$
- $\tau_{C_5} = C_5(R_S||r_\pi + \beta R_{E1}||R_1||R_2)$

Plugging the corresponding values and using $f_l = \frac{\omega_l}{2\pi}$, we obtain:

$$f_l = \frac{1}{2\pi(\tau_{C_4} + \tau_{C_5})} = 13.34\text{Hz}$$

# 3 Results

The values provided were tested on the lab using a breadboard and the oscilloscopes, with an input waveform of 0.1V peak-to-peak, various frequencies and no DC offset. The results can be seen below for the measured values for the DC biasing voltages:

| Quantity | Expected Value | Measured Value |
|---|---|---|
| $V_{B1}$ | 3.1V | 3.11V |
| $V_{E1}$ | 2.4V | 2.44V |
| $V_{E2}$ | 1.7V | 1.72V |
| $V_{C2}$ | 2.9V | 2.93V |
| $V_{B3}$ | 3.6V | 3.64V |
| $V_{C3}$ | 8.6V | 8.58V |

This results in the following current values at DC bias using $I = \frac{V}{R}$ and/or the multimeter:

| Quantity | Expected Value | Measured Value |
|---|---|---|
| $I_{C2}$, $I_{C3}$ | 4.4mA | 4.41mA |
| $I_{R3}$ | 0.204mA | 0.203mA |
| $I_{R4}$ | 0.200mA | 0.202mA |
| $I_{B2}$ | 25.9$\mu$A | 25.9$\mu$A |
| $I_{E1}$ | 92.3$\mu$A | 93.8$\mu$A |
| $I_{B1}$ | 0.543$\mu$A | 0.552$\mu$A |
| $I_{R1}$ | 0.213mA | 0.212mA |
| $I_{R2}$ | 0.207mA | 0.207mA |

Then, the small-signal gain was measured using a $v_{in} = 100m$V peak-to-peak sine signal at a frequency of 200kHz. With the 100:1 attenuator and the 253.85 gain of the two-stage amplifier, we can expect a resulting signal $v_{out} = (100m\text{V})\frac{1}{100}(253.85) = 253m$V. The observed peak-to-peak voltage measured at $v_{out}$ has a value even higher, of 303mV (as shown below), satisfying our expectations.

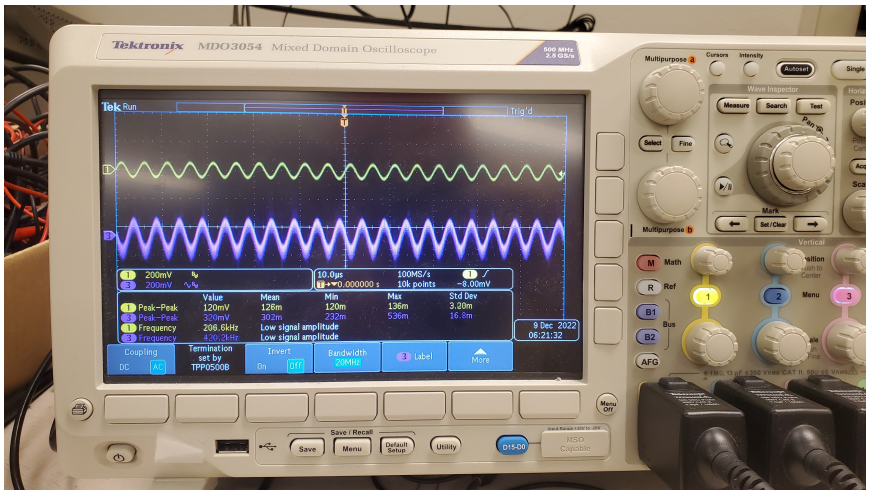



Figure 2: Oscilloscope measurement. The input waveform is of 100mVpp; thus having that the 100:1 attenuator in addition with the gain should return a signal of around 250mVpp

Finally, we measure the frequency response by observing the measured output voltage at frequencies from 100kHz to 20MHz:

| Frequency | Output peak-to-peak voltage |
|---|---|
| 100kHz | 307mV |
| 200kHz | 302mV |
| 300kHz | 306mV |
| 400kHz | 309mV |
| 500kHz | 305mV |
| 600kHz | 302mV |
| 700kHz | 302mV |
| 800kHz | 300mV |
| 900kHz | 304mV |
| 1MHz | 306mV |
| 2MHz | 299mV |
| 3MHz | 284mV |
| 4MHz | 273mV |
| 5MHz | 261mV |
| 6MHz | 246mV |
| 7MHz | 232mV |
| 8MHz | 215mV |
| 9MHz | 198mV |
| 10MHz | 180mV |
| 11MHz | 164mV |
| 12MHz | 152mV |
| 13MHz | 149mV |
| 14MHz | 141mV |
| 15MHz | 137mV |
| 16MHz | 134mV |
| 17MHz | 135mV |
| 18MHz | 138mV |
| 19MHz | 143mV |
| 20MHz | 140mV |

As observed, we do not observe a large drop on the measurements for the gain until we break the 4Mhz line; as expected from our calculations in the lab.